

West Africa's Crime Rate

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Introduction

Crime remains a significant challenge to peace, security, and socioeconomic development across West Africa, with violent conflicts and civil strife long recognised as major destabilising forces in the region (Annan, 2014). Major incidents recorded across multiple countries and cities since 1974. Understanding where crime is most concentrated, and how it varies across regions, is essential for informing policy decisions, resource allocation, and community safety interventions. To read more about crimes in West Africa [Click Here](#)

This report presents an exploratory analysis of a simulated crime dataset covering cities in Nigeria and Cameroon. The objective is to construct the dataset from its constituent variables, classify cities into crime rate categories, and summarise key patterns; including the total number of crimes, the country with the highest crime burden, and the average number of crimes per country.

The remainder of this report is organised below:



Figure 1: A man being robbed in Cameroon

Table 1: Crime counts by city and country

```
country <- c(rep("Nigeria", 3), "Cameroon",  
  rep("Nigeria", 3), "Cameroon",  
  rep("Nigeria", 8), "Cameroon",  
  rep("Nigeria", 3))
```

Analysis Breakdown

This analysis makes use of a combination of **base R** functions and functions from the **dplyr** package to create, manipulate, and summarise the data.

```
library(dplyr)  
library(knitr)
```

Creating the country, city, and n_crime vectors

The `country` vector is created using `rep()`, which repeats each country name the number of times it appears in **Table 1**. The order matches the table, with "Cameroon" placed where it occurs between the Nigeria entries.

`length(country)` confirms the vector has 20 entries, matching the number of cities in Table 1, while `unique(country)` prints the distinct countries represented: Nigeria and Cameroon.

```
length(country)
```

```
[1] 20
```

```
unique(country)
```

```
[1] "Nigeria" "Cameroon"
```

The `city` and `n_crime` vectors are created directly, in the same row order as Table 1, so that the three vectors align correctly when combined later.

```
city <- c("Maiduguri", "Unknown", "Kano", "Bamenda",
        "Port Harcourt", "Potiskum", "Damaturu",
        "Unknown", "Jos", "Lagos", "Gombe", "Kaduna",
        "Logo District", "Konduga", "Warri", "Bama",
        "Buea", "Buni Yadi", "Damboa", "Gamboru"
        )

n_crime <- c(488, 225, 130, 87, 76, 65, 63, 55, 51,
            46, 42, 38, 37, 36, 35, 32, 31, 30, 30, 29)
```

Data types and structures

```
## Data Type
typeof(country)
```

```
[1] "character"
```

```
typeof(city)
```

```
[1] "character"
```

```
typeof(n_crime)
```

```
[1] "double"
```

```
## Data Structure
str(country)
```

```
chr [1:20] "Nigeria" "Nigeria" "Nigeria" "Cameroon" "Nigeria" "Nigeria" ...
```

```
str(city)
```

```
chr [1:20] "Maiduguri" "Unknown" "Kano" "Bamenda" "Port Harcourt" ...
```

```
str(n_crime)
```

```
num [1:20] 488 225 130 87 76 65 63 55 51 46 ...
```

All three objects are **atomic vectors**: `country` and `city` are of type `character`, while `n_crime` is of type `double`. `str()` confirms each is a vector of length 20.

Total number of crimes committed in West Africa

The total number of crime committed is 1626

Storing the data using the function: `data.frame`

Since Table 1 contains multiple variables of different types (character and numeric) organised in rows and columns, the appropriate data structure to store it is a **data frame**, created here using the three vectors above.

```
crime_tbl <- data.frame(  
  country,  
  city,  
  n_crime  
)  
  
crime_tbl |> knitr::kable()
```

country	city	n_crime
Nigeria	Maiduguri	488
Nigeria	Unknown	225
Nigeria	Kano	130
Cameroon	Bamenda	87
Nigeria	Port Harcourt	76
Nigeria	Potiskum	65
Nigeria	Damaturu	63
Cameroon	Unknown	55
Nigeria	Jos	51
Nigeria	Lagos	46
Nigeria	Gombe	42
Nigeria	Kaduna	38

country	city	n_crime
Nigeria	Logo District	37
Nigeria	Konduga	36
Nigeria	Warri	35
Nigeria	Bama	32
Cameroon	Buea	31
Nigeria	Buni Yadi	30
Nigeria	Damboa	30
Nigeria	Gamboru	29

Classifying crime rate

Two methods are used to classify each city's `n_crime` into **High**, **Medium**, or **Low**, based on Table 2.

Method 1: Using Base R `ifelse()`

```
crime_tbl$crime_rate <- ifelse(
  test = crime_tbl$n_crime >= 400,
  yes = "High",
  no = ifelse(
    test = between(crime_tbl$n_crime, 100, 399),
    yes = "Medium",
    no = "Low"
  )
)

crime_tbl |> knitr::kable()
```

country	city	n_crime	crime_rate
Nigeria	Maiduguri	488	High
Nigeria	Unknown	225	Medium
Nigeria	Kano	130	Medium
Cameroon	Bamenda	87	Low
Nigeria	Port Harcourt	76	Low
Nigeria	Potiskum	65	Low
Nigeria	Damaturu	63	Low
Cameroon	Unknown	55	Low
Nigeria	Jos	51	Low
Nigeria	Lagos	46	Low

country	city	n_crime	crime_rate
Nigeria	Gombe	42	Low
Nigeria	Kaduna	38	Low
Nigeria	Logo District	37	Low
Nigeria	Konduga	36	Low
Nigeria	Warri	35	Low
Nigeria	Bama	32	Low
Cameroon	Buea	31	Low
Nigeria	Buni Yadi	30	Low
Nigeria	Damboa	30	Low
Nigeria	Gamboru	29	Low

Method 2: Loading dplyr and using if_else()

```

crime_tbl$crime_rate <- if_else(
  condition = crime_tbl$n_crime >= 400,
  true = "High",
  false = if_else(
    condition = between(crime_tbl$n_crime, 100, 399),
    true = "Medium",
    false = "Low"
  )
)

crime_tbl |> knitr::kable()

```

country	city	n_crime	crime_rate
Nigeria	Maiduguri	488	High
Nigeria	Unknown	225	Medium
Nigeria	Kano	130	Medium
Cameroon	Bamenda	87	Low
Nigeria	Port Harcourt	76	Low
Nigeria	Potiskum	65	Low
Nigeria	Damaturu	63	Low
Cameroon	Unknown	55	Low
Nigeria	Jos	51	Low
Nigeria	Lagos	46	Low
Nigeria	Gombe	42	Low
Nigeria	Kaduna	38	Low
Nigeria	Logo District	37	Low

country	city	n_crime	crime_rate
Nigeria	Konduga	36	Low
Nigeria	Warri	35	Low
Nigeria	Bama	32	Low
Cameroon	Buea	31	Low
Nigeria	Buni Yadi	30	Low
Nigeria	Damboa	30	Low
Nigeria	Gamboru	29	Low

Both methods produce identical results. `ifelse()` is base R and needs no extra package, while `if_else()` (from `dplyr`) is stricter about type matching between `true` and `false`, which helps catch errors earlier.

Summary statistics

Total and average number of crimes per country

```
crime_tbl |>
  summarise(
    .by = country,
    total_crime = sum(n_crime),
    average_number_of_crime = mean(n_crime)
  ) |>
  knitr::kable()
```

country	total_crime	average_number_of_crime
Nigeria	1453	85.47059
Cameroon	173	57.66667

Country with the highest crime committed

```
crime_tbl |>
  summarise(
    .by = country,
    total_crime = sum(n_crime),
  ) |>
  top_n(n = 1,
        wt = total_crime) |>
  knitr::kable()
```

country	total_crime
Nigeria	1453

Annan, N. (2014). Violent conflicts and civil strife in west africa: Causes, challenges and prospects. *Stability*, 3(1). <https://doi.org/10.5334/sta.da>